

Starbucks Nutrition Insights Business Report

Power BI Consumer Analytics Dashboard

Project type	Power BI consumer analytics dashboard
Dataset	Starbucks nutritional information
Focus	Calories, caffeine, sugar, fat, carbohydrates, and dairy/non-dairy choices

1. Executive Summary

This project presents a real-world Power BI dashboard that analyses the nutritional profile of Starbucks beverages. The purpose of the dashboard is to help customers make more informed drink choices by comparing caffeine, calories, sugar, carbohydrates, fat, cholesterol-labelled values, and dairy/non-dairy options across beverage categories.

The dataset contains 242 beverage variations across 9 beverage categories. The analysis shows that brewed coffee is the strongest caffeine option, averaging approximately 294 mg of caffeine. Smoothies and Frappuccino-style drinks are the most calorie-heavy categories, while Frappuccino Blended Coffee also shows the highest average sugar level at approximately 57 g. Dairy and non-dairy options both appear across the menu, but non-dairy should not automatically be treated as lower calorie or lower sugar; customers still need to compare the full nutritional profile.

Overall, the dashboard supports healthier and more transparent beverage decisions by turning raw menu nutrition data into practical consumer insights.

2. Business Scenario

Starbucks customers often choose drinks based on taste, habit, or brand familiarity, but many drinks vary significantly in caffeine, calories, sugar, and milk preparation. This creates a challenge for customers who are health-conscious, lactose-intolerant, caffeine-sensitive, or trying to manage daily sugar intake.

The dashboard is designed from the perspective of a retail analytics or consumer insights team. Its goal is to convert nutritional data into a simple decision-support tool that helps customers quickly compare beverage options and identify drinks that better match their lifestyle or dietary preferences.

3. Data Preparation Summary

The raw Excel data was prepared in Power BI using Power Query before creating the dashboard. The main preparation steps included:

- Removing unnecessary metadata rows and keeping the actual header row.
- Standardising column names to make nutrition fields easier to understand.
- Converting numeric fields such as calories, caffeine, sugars, carbohydrates, fat, and protein into appropriate number formats.
- Handling missing or unclear values so visuals would not break or mislead the analysis.
- Creating a dairy classification field to group drink preparations into Dairy, Non-Dairy, and Other/Unspecified categories.
- Organising the data so that beverage category, beverage name, and milk preparation could be used cleanly as filters and slicers.

These preparation steps improved the reliability of the dashboard and made the visuals easier to interpret.

4. Dashboard Analysis and Key Insights

Insight Area	Key Finding	Consumer Meaning
Caffeine	Coffee has the highest average caffeine level at approximately 294 mg. Classic espresso drinks are the next highest major category at around 122 mg.	Customers who want a strong caffeine drink may prefer brewed coffee, while caffeine-sensitive customers should be careful with coffee-based options.
Calories	Smoothies and Frappuccino Blended Coffee are the highest calorie categories, averaging around 282 and 277 calories respectively.	These drinks are better treated as indulgent or occasional choices rather than everyday light beverages.
Sugar	Frappuccino Blended Coffee has the highest average sugar level at approximately 57 g, followed by Frappuccino Blended Creme at around 48 g.	Customers managing sugar intake should compare Frappuccino-style drinks carefully before ordering.
Dairy and non-dairy	The dataset contains approximately 149 dairy preparations, 66 non-dairy preparations, and 27 other/unspecified options.	Non-dairy options are useful for lactose-intolerant or plant-based customers, but they should still be checked for calories and sugar.
Fat and cholesterol-labelled values	Milk-heavy drinks generally show higher fat-related values than simple brewed coffee or tea options, although the relationship is not perfectly linear.	Customers concerned about fat intake may prefer lighter drinks such as brewed coffee, tea, or selected shaken beverages.

5. Consumer Recommendations

- Choose brewed coffee or tea when looking for a lighter beverage option.
- Review Frappuccino and smoothie options carefully because they often contain higher calories and sugar.
- Use caffeine comparisons before choosing larger coffee-based drinks, especially for customers who are caffeine-sensitive.
- Treat non-dairy options as lactose-friendly alternatives, not automatically as low-calorie choices.
- Compare both beverage category and preparation type, because milk choice and size can significantly change the nutritional profile.

6. Business Value of the Dashboard

This dashboard can support both customers and business teams. For customers, it provides a clearer way to compare drinks and make choices aligned with health goals. For business teams, it can support menu transparency, product positioning, customer segmentation, and health-focused communication.

From a portfolio perspective, this project demonstrates practical Power BI skills including data cleaning, data transformation, visual storytelling, dashboard design, and consumer-focused insight generation.

7. Limitations

This analysis is based on the available Starbucks nutrition dataset and may not reflect the most current Starbucks menu or regional product variations. The dataset does not include pricing, sales volume, customer demographics, or order frequency, so it cannot show which drinks are most profitable or most popular. Some nutrition field labels should also be validated against official Starbucks definitions before using this report for operational or medical decision-making.

8. Conclusion

The Starbucks Nutrition Insights Dashboard shows how Power BI can turn raw nutritional information into a practical decision-making tool. The strongest insights relate to caffeine differences, calorie-heavy beverage categories, sugar concentration in blended drinks, and the availability of dairy and non-dairy alternatives.

The project highlights that healthier choices are not always obvious from the drink name alone. A well-designed dashboard can help customers compare options more clearly and support better everyday beverage decisions.